

HSE Intelligence — Mobile Application Architecture

4 Role-Based Apps · 10 Operational Workflows · React Native 0.85 + FastAPI + MySQL

Rev 2.0 — post client review, 30 Jul 2026
Aligned to: HSE Master Data & Forms v1.0 (S. Sheikh)

6 BUILT

4 GAPS

LAYER 1 · MOBILE APPLICATION SHELL — shared across all four role apps

Auth & JWT Login, refresh, force password change	RBAC Role Router Backend role is authoritative	API Client Axios + interceptors + retry	GPS Capture Lat/long on all reports	Camera / Evidence Photo + attachment upload	QR / Barcode Scan Badge, asset, vehicle ID	Offline Queue Draft & sync on reconnect
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WORKER Employee · Field Operative src/worker/ Report Incident Type, severity, injury & body part, control failure, GPS, hazard link, photosWF-03 Near Miss / Unsafe Act Consequence, cause, hazard-still-present, GPSWF-03 Report Risk / Hazard Category, severity × probability, existing controlsWF-01 Request Permit Type, work description, station, validity windowWF-02 Safety Checklist & Tasks Assigned tasks, checklist submission, shift check-inWF-05 My Training & Competency Training matrix, e-learning, assessment score, cert expiry alerts, toolbox sign-offWF-07 Journey Management (JMP) Vehicle QR scan, fatigue hours declaration, pre-trip check, route check-in, arrival confirmWF-10 My Safety Score Personal PIRS contribution & training currency indicatorWF-08	SUPERVISOR Line Manager · Area Authority src/screens/ Incident Investigation Acknowledge, 5-Whys RCA, days away, regulatory notified, signatureWF-03/04 CAPA Assignment Action description, owner, due date, escalate to ManagerWF-04 Permit Acknowledgement Review, supervisor notes, actual work times, deviation flagWF-02 Safety Walk / Inspection Issues found, housekeeping & compliance rating, follow-upWF-05 Team & Shift Management Attendance, shift monitoring, toolbox talks, task assignmentWF-06 Team Training Matrix Competency gaps by role, expiring certs, book & verify trainingWF-07 Authorise Journey JMP approval signature, live check-in monitor, overdue escalationWF-10 Contractor Site Control Induction verify, permit-to-access, contractor worker roll-callWF-09	SAFETY MANAGER HSE Manager · Site Owner src/manager/ Incident Close-out Manager queue, approve investigation, lessons learned, closureWF-03 Permit Approvals Approve / reject with reason, active permit monitoringWF-02 CAPA Compliance Sign-off Validate closure evidence, mark completeWF-04 Hazard Register Review Open hazards → under review → controlled, risk heat mapWF-01 Monitoring & Complaints Live site KPIs, audit trail, complaint triageWF-06 Safety Performance Score SPS dashboard, KPI weights & targets, RAG thresholds, PIRS trendWF-08 Contractor Pre-Qualification 3-yr TRIR/LTIFR history, scoring, approve site access, suspendWF-09 Fleet & Fatigue Oversight Fatigue breaches, weather/route risk, vehicle compliance expiryWF-10 Competency Approval Sign off competency, authorise permit authority levelWF-07	AUDITOR Safety Officer · Independent Assurance src/auditor/ Assigned Audits Scheduled / in-progress / overdue, site & department scopeWF-05 Audit Checklist Execution Pass / fail / N-A per item, remarks, photo evidence, scoreWF-05 Permit Verification On-site check: valid / not displayed / invalidWF-02 Hazard Verification Confirm hazard is being actively managed on siteWF-01 Audit Calendar Forward schedule & compliance evidence trailWF-06 Training Record Audit Verify certificates against competency matrix, flag expiredWF-07 Contractor Audit On-site contractor compliance score & corrective findingsWF-09 Vehicle & JMP Audit Pre-trip inspection sampling, licence & roadworthiness checkWF-10
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LAYER 2 · API GATEWAY — FastAPI, JWT + org-scoped RBAC on every route

Incidents /incident-workflow/* /near-miss-workflow/* /unsafe-act-workflow/*	Permits /permit-workflow/* /permit-to-works/*	Risk & Hazards /risk-workflow/* /hazard-register/*	CAPA & Audit /capa-actions/* /audits/* · /safety-walks/*	Training △ NEW /training-records/* /competency-matrix/*	Transport △ NEW /journey-plans/* /vehicles/* · /fatigue/*
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LAYER 3 · WORKFLOW STATE ENGINE — Worker → Supervisor → Manager → Auditor state machine per workflow

WF-01 Risk Assessment BUILT	WF-02 Permit to Work BUILT	WF-03 Incident Reporting BUILT	WF-04 CAPA BUILT	WF-05 Audit & Inspection BUILT	WF-06 Compliance Trail BUILT	WF-07 Training & Competency GAP · LOW	WF-08 Safety Perf. Score GAP · LOW	WF-09 Contractor Mgmt GAP · MED	WF-10 Transport / JMP GAP · HIGH
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LAYER 4 · AI INTELLIGENCE ENGINE — advisory only; the engine recommends, humans decide

Policy & KPI Gap Analyser Ingests policies, objectives, registers → reports KPI & SOP compliance gaps	Risk Identifier + Heat Map Classifies historical hazards → risk register + 5×5 red/amber/green matrix	KPI Generation & Performance TRIR · LTIFR · DART · FAR · near-miss ratio · CAPA closure	Predictive Injury Risk Score PIRS Per employee / area / site — fatigue, training currency, incident history
Corrective Action Suggester Engineering · administrative · procedural · supervision · training controls	Control of Work Intelligence Permit risk indicators, tightening recommendations, real-time elevated-risk alerts	Safety Communication Analysis Toolbox talks, briefings & comms coverage opinion	Continuous Learning Loop NEW Controls, overrides & benchmarks feed back into model improvement

LAYER 5 · DATA LAYER — MySQL · master data is referenced by every operational record (cascade & immutable audit trail)

MASTER DATA — Organisation → BU → Region → Country → Site → Department → Work Area → Employee → Role OrganisationSiteDepartmentWork AreaEmployeeRole / Job ProfileEquipmentPermit TypeHazard CategoryShift PatternVehicle △ Contractor Company △Contractor Worker △Training Programme △Certification Type △Hazardous Substance △	OPERATIONAL DATA — transactional records created by daily HSE activity IncidentsNear MissesUnsafe ActsPermitsCAPA ActionsAuditsSafety WalksHazard RegisterChecklists Shift SchedulesJourney Plans △Training Records △Contractor Scores △SPS Snapshots △
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■ Built & backend-integrated ■ Gap — raised in client review ■ New capability

Roles map to client access matrix: Employee → Worker · Supervisor → Supervisor · Safety Officer → Auditor · Safety Manager → Manager · (Safety Director & System Admin = web console)